Aditi Nair

addynair6@gmail.com • linkedin.com/in/justaddy • github.com/addynair • aditinair.vercel.app

Profile / Summary

Electronics and Computer Engineering undergraduate with hands-on experience building full-stack web applications, REST API backends, and AI-powered systems using Python, C++, TypeScript, and Node.js. Comfortable working on Linux-based environments with strong Git-based version control habits and clean, modular code practices. Experienced in cross-functional collaboration on open-source and academic projects. Eager to contribute to backend development, scripting, and automation tasks in a fast-paced engineering team.

Educational Qualifications

Bachelor of Technology in Electronics and Computer Engineering
Amrita Vishwa Vidyapeetham, Kerala, India — Aug 2023 – Present

Secondary and Senior Secondary Education (Class 10th & 12th)
Amrita Vidyalayam, Kerala — 2013 – 2023

Technical Skills

Languages: C++, Python, JavaScript, TypeScript, SQL
Web & Backend: Node.js, Express.js, REST APIs, Next.js, React.js, React Native, HTML, CSS, Tailwind CSS
Databases: PostgreSQL, MongoDB, MySQL
ML/AI: LangChain, Hugging Face Models, Groq API (LLaMA 3.3 70B), scikit-learn, OpenCV, MediaPipe
DevOps & Tools: Git, GitHub, Vercel, Render, Figma, Neon (cloud PostgreSQL)
Auth & Security: JWT, HTTP-only Cookies, Middleware-based Route Protection
Cloud / Infra: Microservices architecture, REST API design, cloud deployment (Vercel, Render); exposure to distributed systems concepts through coursework
DSA: Solved 80+ LeetCode problems (Easy–Medium); strong in arrays, two pointers, hashing, and binary search

Projects

CollabCode – Real-Time Collaborative Cloud IDE *Ongoing*
Node.js, Socket.IO, Monaco Editor, MongoDB Atlas, GitHub OAuth, JWT, React.js

- Built a full-stack real-time collaborative coding platform supporting multi-user shared sessions via unique invite links, with instant code synchronization powered by Socket.IO WebSockets.
- Integrated Monaco Editor (the engine behind VS Code) to deliver an IDE-like experience with syntax highlighting, minimap navigation, and multi-language support across JavaScript, TypeScript, Python, HTML, CSS, and JSON.
- Designed a complete multi-file workspace with live file operations synchronized across all connected collaborators in real time, including live presence tracking and collaborative cursor synchronization.
- Implemented an event-driven backend architecture on Node.js with MongoDB Atlas for persistent storage of sessions, users, and file contents – ensuring state survives disconnects and page refreshes.
- Built GitHub OAuth and JWT-based authorization for secure login and protected session access, following authentication best practices.

CeeVee – Resume Skill Gap Analyzer *March 2026*
Next.js 16, TypeScript, Node.js, PostgreSQL, Prisma, Groq API, Fuse.js, Recharts, JWT, Vercel, Neon

- Designed and deployed a full-stack REST API-backed web application using Node.js and PostgreSQL.
- Built a multi-layer file validation and parsing pipeline (PDF/DOCX) to detect and reject invalid inputs before AI processing – demonstrating scripting-style defensive programming and automation thinking.
- Integrated Groq LLaMA 3.3 70B via API for intelligent skill extraction, STAR-method bullet rewriting, and personalized learning roadmap generation, showcasing practical AI/ML toolchain integration.
- Implemented a fuzzy-search JD matching engine (Fuse.js) to compute skill match percentages and priority-rank missing skills, reflecting algorithmic problem-solving with real-world data.
- Developed JWT-based authentication with HTTP-only cookies and middleware-based route protection, following secure software engineering best practices.
- Deployed on Vercel with a Neon cloud PostgreSQL database and integrated Vercel Analytics for usage observability.

Strumly – ML-Powered Air-Guitar Chord Recognition System *Nov 2025*
Python, OpenCV, scikit-learn, Random Forest, MediaPipe, Git

- Built a real-time chord recognition system in Python using MediaPipe for hand landmark tracking and a Random

Forest classifier trained on a custom dataset, achieving 94% classification accuracy.
- Developed computer vision pipelines with OpenCV and trained/evaluated ML models using scikit-learn, demonstrating strong Python scripting and data pipeline skills.
- Collaborated with a teammate via Git for version control, branching, and code review – practising engineering best practices in a team environment.

Leadership & Open-Source Experience

Member, amFOSS (Free and Open Source Software Community) Nov 2023 – Feb 2025

- Collaborated with a team of 30+ developers on open-source projects using Git and GitHub workflows – daily practice of pull requests, branching, and code review aligned with engineering team norms.
- Led Figma-based UI/UX design for Hacktoberfest events organized by amFOSS, coordinating across design, development, and event teams.
- Contributed to ICPC website redesign using HTML, CSS, and JavaScript alongside faculty mentors, demonstrating ability to execute tasks with guidance and take proactive ownership.

Social Student Responsibility (SSR) Initiative Nov 2025

- Delivered a session titled "School to Society: Journey towards Inclusion and Empathy" for 35 students at a government school in Trivandrum, demonstrating strong communication and public speaking skills.
- Coordinated with a team to plan and execute the initiative as part of a college-led social responsibility programme.

Certifications

Google Foundations of UX Design (Coursera) Aug 2024
Covered topics: user-centered design, accessibility, usability testing, and design systems.

Area of Interest

Backend Systems & Web Technologies, Artificial Intelligence, Machine Learning, Mobile Development, Distributed Systems

Soft Skills & Additional Information

Soft Skills: Communication, Teamwork, Adaptability, Proactive Learning, Problem-solving
Languages: English (Fluent), Hindi (Native), Malayalam (Proficient)
Hobbies: Reading, Sports, Writing